CIS 3252W.E01 (Su26-OU1st5wk)
Business Intelligence

WHY DO SEOUL'S BIKES STAY BUSY?

Final Business Intelligence Analysis Using Tableau

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THE PROBLEM

- Weather
- Seasons
- Rush hours
- Holidays

Unpredictable demand causes shortages, overcrowded stations, and operational inefficiencies.

THE DATASET

Dataset Overview

Seoul Bike Sharing Dataset
UCI Dataset 560
8,760 hourly observations
December 2017 –
November 2018

Key Variables

Temperature
Rainfall
Snowfall
Seasons
Holiday
Hour
Rented Bike Count

About the Data

Hourly bike rental records collected across Seoul over a full calendar year.

Captures seasonal variation, weather conditions, and commuting patterns to support demand forecasting and operational planning.

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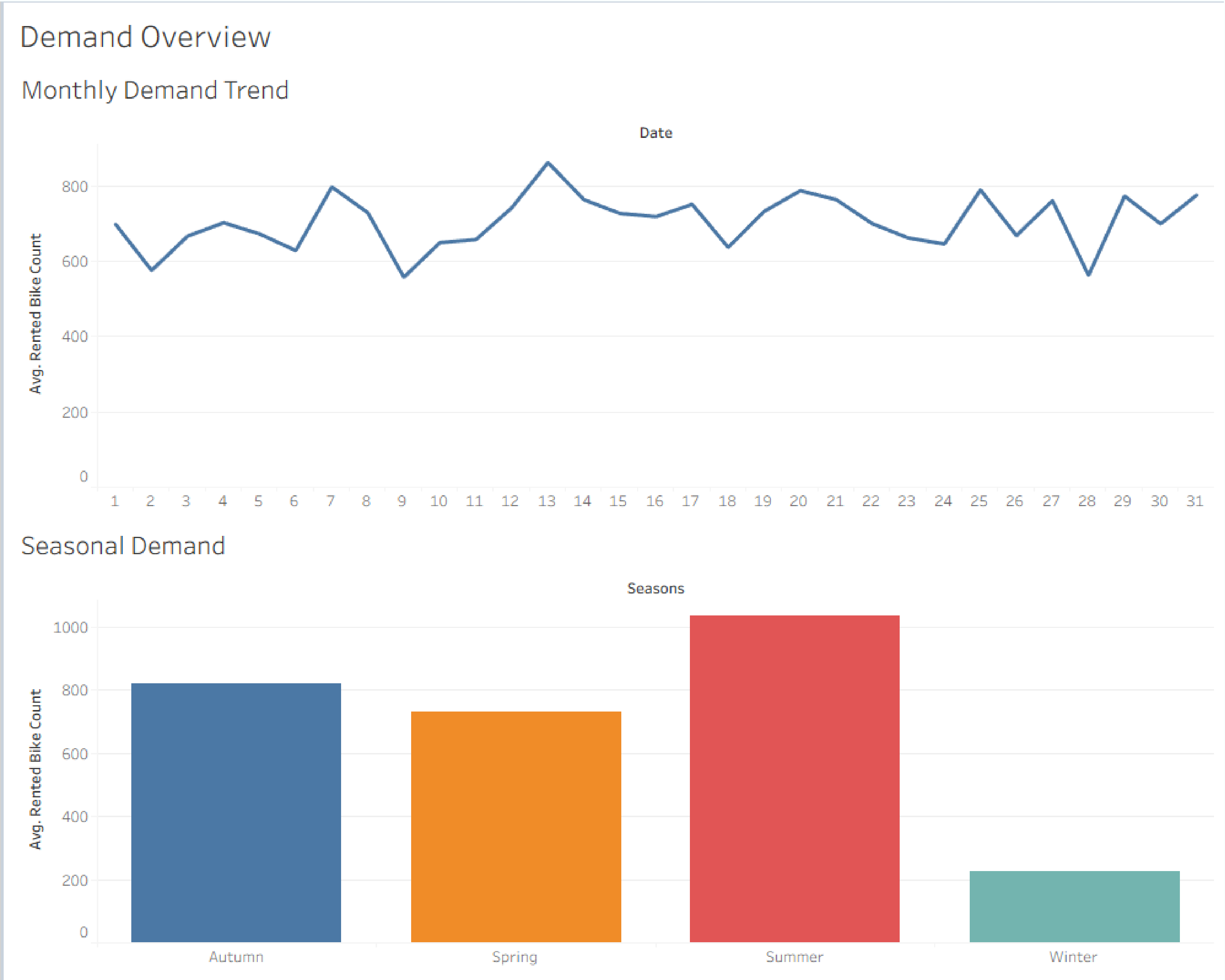
DASHBOARD 1 – DEMAND OVERVIEW



Summer had the highest demand.
Winter had the lowest demand.
Demand remained relatively consistent throughout the year.

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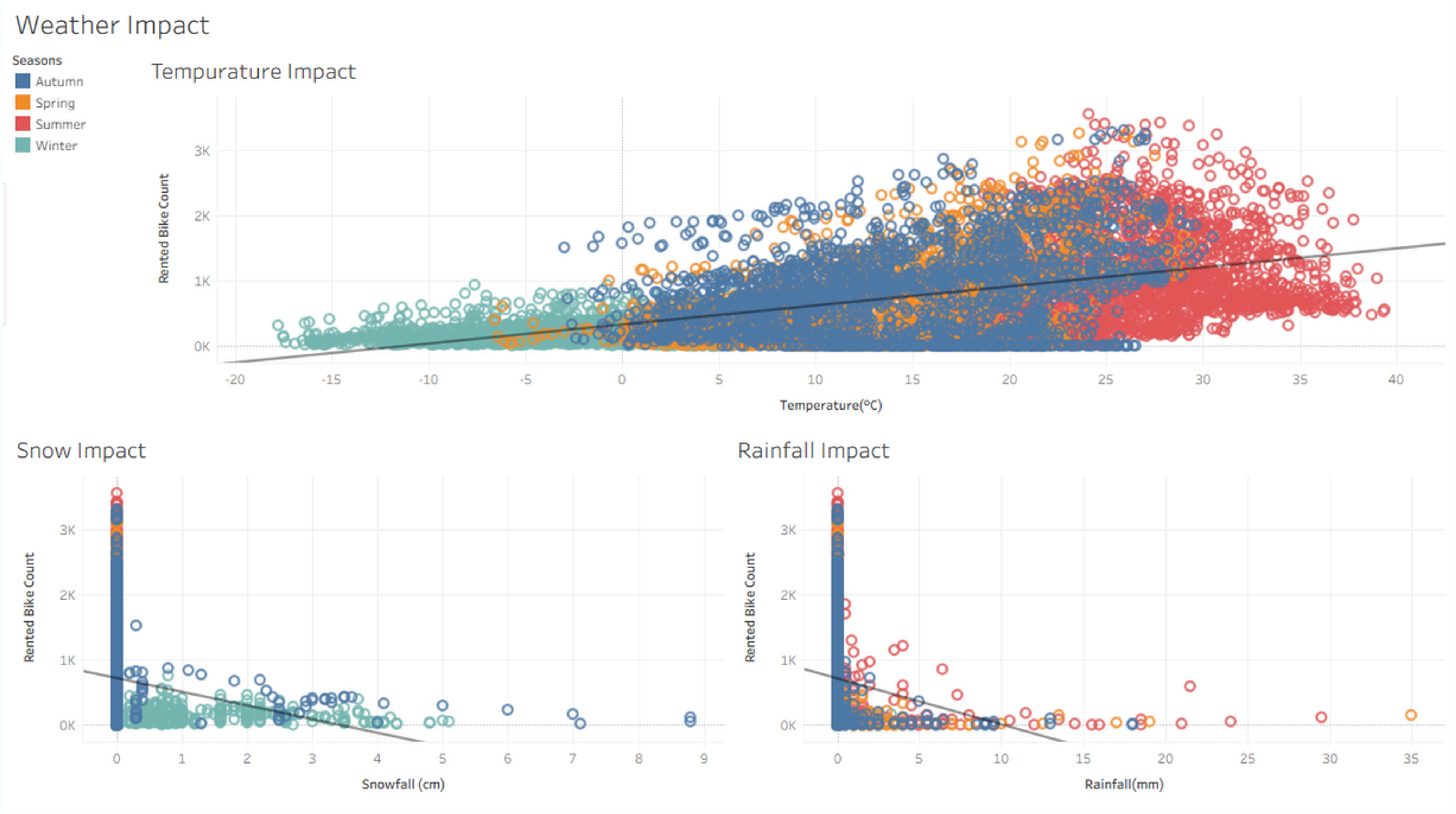
DASHBOARD 2 – WEATHER IMPACT



Temperature
➡ Higher temperatures = more rentals

Rainfall
⬇ Rain reduces demand

Snowfall
⬇⬇ Snow has the strongest negative effect

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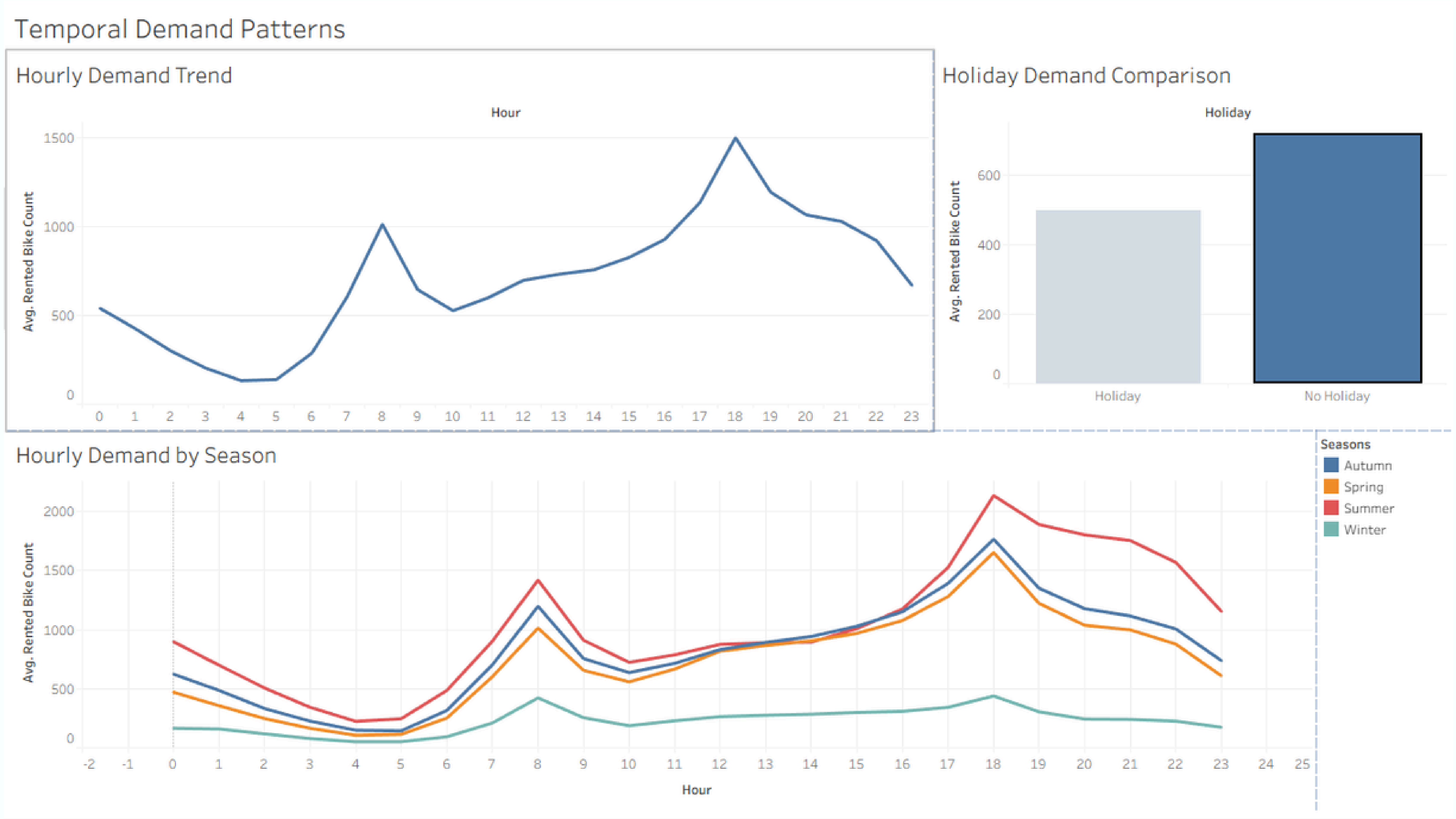
DASHBOARD 3 – TEMPORAL PATTERNS



Morning peak ≈ 8 AM
Evening peak ≈ 6 PM

Summer highest

Holiday demand lower

WHAT DID WE LEARN?

🚲 Fleet Redistribution
Optimize bike distribution to meet demand hotspots.

🌦️ Weather Forecasting
Incorporate weather data for better demand prediction.

📅 Seasonal Planning
Adjust operations according to seasonal usage patterns.

🏗️ Infrastructure
Improve station capacity and locations.

📊 Business Intelligence
Use Tableau dashboards for actionable insights.

✅ Key Finding
Data-driven decisions reduce shortages and inefficiencies.

FINAL TAKEAWAYS

- Weather strongly influences bike demand.
- Demand follows predictable commuting patterns.
- Tableau transformed raw data into actionable business insights.